

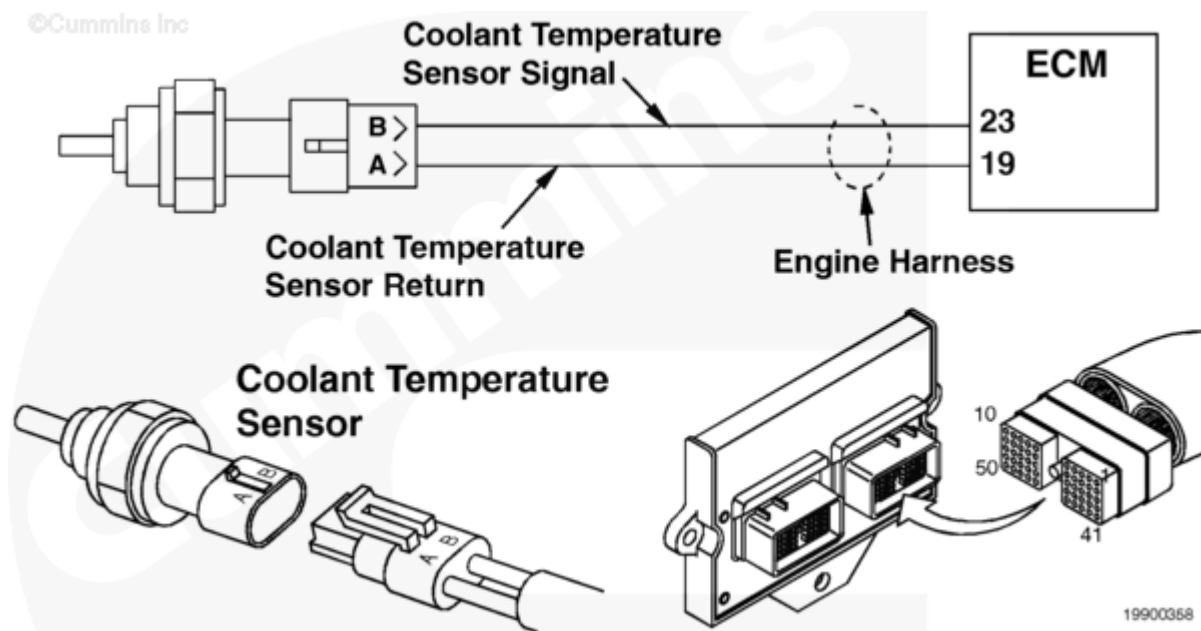
Fault Code: 151

Coolant Temperature - Engine Protection

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Overview

Codes	Reason	Effect
Fault Code: 151 PID(P): P110 SPN: 110 FMI: 0 Lamp: Engine Protection SRT:	High coolant temperature has been detected. Voltage signal at coolant temperature signal pin 23 indicates the coolant temperature is above the calibrated value 100°C [212°F].	Calibration-dependent progressive power and speed derate and engine shutdown as temperature increases over thresholds.



Coolant Temperature Sensor Circuit

Circuit Description

The coolant temperature sensor is used by the electronic control module (ECM) to monitor the temperature of the engine coolant. The coolant temperature is used by the ECM for the engine protection system, timing, and fueling control.

The ECM monitors the voltage on pin 23. The ECM expects to see the voltage vary between 0.5 and 4.5 VDC.

Component Location

The coolant temperature sensor is located on the left side of the engine in the thermostat housing.

Shoptalk

The signal voltage varies between 0.5 and 4.5 VDC as the internal resistance of the sensor changes due to changing coolant temperature. When the sensor signal voltage indicates a temperature exceeding a set limit, Fault Code 151 is logged.

- Make sure the airflow through the radiator is **not** obstructed.
- The resistance of all the temperature sensors varies with the temperature. The reading you observe should compare to the following table if the sensor is functioning properly.

Temperature (°C)	Temperature [°F]	Resistance (ohms)
0	32	30k to 36k
25	77	9k to 11k
50	122	3k to 4k
75	167	1350 to 1500
100	212	600 to 675

Note: The number of fault lamps could be reduced to two for certain OEMs. The engine protection and stop lamps are wired together as a red lamp. The warning lamp remains a yellow lamp.

Refer to Troubleshooting Fault Code t05-151 (</qs3/pubsys2/xml/en/procedures/87/87-t05-151.html>)